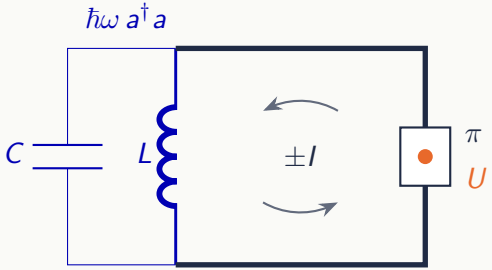
A π loop in a cavity: spontaneous persistent current, exactly

Quantum-dot Josephson loop sharing its inductance with an LC mode, solved by DMRG

Group seminar

The circuit

The circuit



- A superconducting loop closed by a π junction: a quantum dot with one unpaired electron (doublet), energy $+E_J \cos\theta$ instead of $-E_J \cos\theta$.
- The loop *shares* its inductance L with a microwave LC mode of impedance $Z = \sqrt{L/C}$ (galvanic coupling; the inductively coupled case is discussed later).
- No external flux.

Question: does a persistent current appear spontaneously, and what is the quantum state of the mode?

The circuit Hamiltonian

The flux of the mode is an operator,

$$\Phi = \Phi_{\mathrm{zpf}}\,(a + a^\dagger), \qquad \Phi_{\mathrm{zpf}} = \sqrt{\hbar Z/2},$$

and the Cooper-pair phase across the junction is $\theta = 2\pi\Phi/\Phi_0 \equiv 2g\,(a + a^\dagger)$:

$$H = \hbar\omega\, a^\dagger a + E_J \cos\big[2g\,(a + a^\dagger)\big]$$

$g^2 = \dfrac{\pi}{4}\dfrac{Z}{R_Q}$, $R_Q = h/4e^2 = 6.45\,\mathrm{k\Omega}$:
$Z = 50\,\Omega \to g = 0.08$, $Z = 1\,\mathrm{k\Omega} \to g = 0.35$.

In the phase θ and its conjugate charge n this is an rf-SQUID (a fluxonium with a π element):

$$H = 4E_C\, n^2 + \tfrac{1}{2}E_L\,\theta^2 + E_J \cos\theta,$$

$$E_L = \frac{\hbar\omega}{8g^2}, \qquad E_C = \hbar\omega g^2.$$

- Two knobs: $\beta = \dfrac{E_J}{E_L} = \dfrac{8g^2 E_J}{\hbar\omega} = \dfrac{2\pi L I_c}{\Phi_0}$ and $E_J/\hbar\omega$.
- $\theta = 0$ is a *maximum* of $E_J \cos\theta$: a negative inductance. For $\beta > 1$ it beats the mode's $\frac{1}{2}E_L\theta^2$ and the potential becomes a double well.

Galvanic or inductive coupling? What the one-mode model assumes

Loop inductively coupled to a detached LC circuit – two modes, the loop's own flux mode b and the resonator a:

$$H = E_C N^2 + E_L\left(\varphi - g_r(a + a^\dagger)\right)^2 + E_J \cos\varphi + \hbar\omega_r a^\dagger a$$

φ: pair phase of the junction; $E_L = (\Phi_0/2\pi)^2/2L_r(1-k^2)$: loop inductance screened by the resonator;
$g_r(a + a^\dagger) = 2\pi M I^{\mathrm{res}}/\Phi_0$: resonator flux through the loop;
$k = M/\sqrt{L_r L_c}$.

Static threshold: minimize over the resonator flux

$$\frac{1}{E_L^{\mathrm{eff}}} = \frac{1}{E_L} + \frac{4g_r^2}{\hbar\omega_r} \;\Leftrightarrow\; L_{\mathrm{eff}} = L_r(1-k^2) + \frac{M^2}{L_c} = L_r.$$

$\beta = 2\pi L_r I_c/\Phi_0$ is set by the loop's *own* inductance; the detached resonator does not lower it, it is *displaced* ($\langle a + a^\dagger\rangle \propto M I_r$) once the current flows.

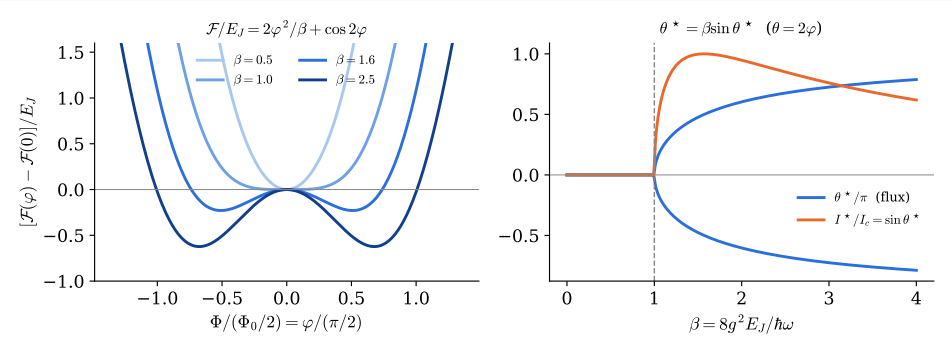
One-mode model $= k \to 1$

All of L_r is the resonator inductor (galvanic, fluxonium-like): $E_L \to \infty$ locks
$\varphi = g_r(a + a^\dagger)$,
$H \to \hbar\omega a^\dagger a + E_J \cos[2g(a + a^\dagger)]$,
$\beta = 8g^2 E_J/\hbar\omega = 2\pi L I_c/\Phi_0$; the loop mode b ($\omega_{rf} = 2\sqrt{E_C E_L}$) is frozen out.

What changes for $k < 1$

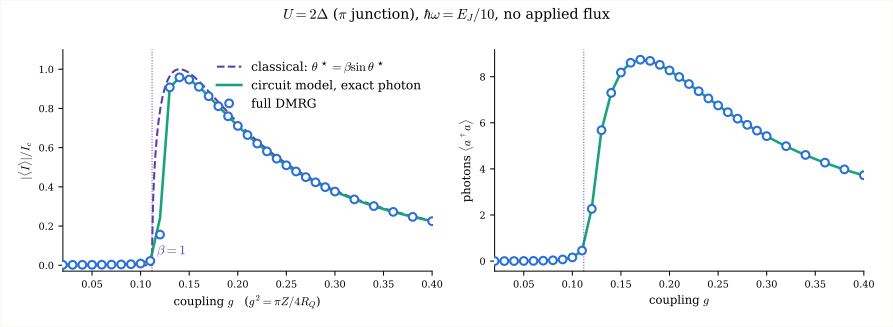
Same classical threshold ($L \to L_r$) and cat physics, but two zero-point fluxes: δ, photon numbers and the flux-driven 0–π shift are shared between a and b (second bosonic site in DMRG; not yet computed).

Classical picture: a double well and a pitchfork



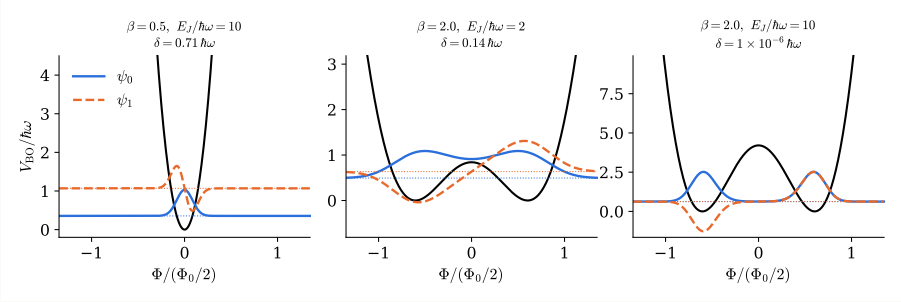
Minima at $\theta^\star = \beta \sin\theta^\star$ carry the current $I^\star = I_c \sin\theta^\star$ and a static flux $\langle\Phi\rangle = LI^\star$: a "photon condensate" $\langle a\rangle \neq 0$. The exact identity $\langle I\rangle = \langle\Phi\rangle/L$ holds in every quantum eigenstate.

The persistent current that emerges



- Above $g_c \simeq 0.13$: a current of $0.96\, I_c$ and a condensate of 7–9 photons. The threshold lies slightly above the classical $\beta = 1$; beyond it the current falls as $\hbar\omega/g^2$ (the flux saturates at $\Phi_0/2$).
- Circles: full microscopic solution (next part); line: circuit Hamiltonian with an exact photon – they agree to 0.2% of I_c.

What is the quantum ground state? A cat



Exact eigenstates of the circuit Hamiltonian. For $\beta > 1$ the ground state is the symmetric superposition $(|\circlearrowleft\rangle + |\circlearrowright\rangle)/\sqrt{2}$ of the two current directions, split from its partner by a tunnel gap $\delta \sim \hbar\omega\, e^{-(E_J/\hbar\omega)\, s(\beta)}$.

A tiny flux offset ($\delta\Phi > \delta/2I^\star$) selects one direction. If $E_J \lesssim \hbar\omega$ the well is shallower than the zero-point energy and no current state exists at all (middle panel).

Where does $E_J \cos\theta$ come from? The microscopic model

The microscopic model: a quantum dot in a BCS ring, plus the mode

$$H = \hbar\omega\, a^\dagger a \underbrace{- t \sum_{r,\sigma} \left(c^\dagger_{r+1\sigma} c_{r\sigma} + \text{h.c.}\right) + \Delta \sum_{r} \left(c^\dagger_{r\uparrow} c^\dagger_{r\downarrow} + \text{h.c.}\right)}_{\text{BCS ring, } L-1 \text{ sites}} + \underbrace{\epsilon_d\, n_0 + U\, n_{0\uparrow} n_{0\downarrow}}_{\text{Anderson dot}}$$

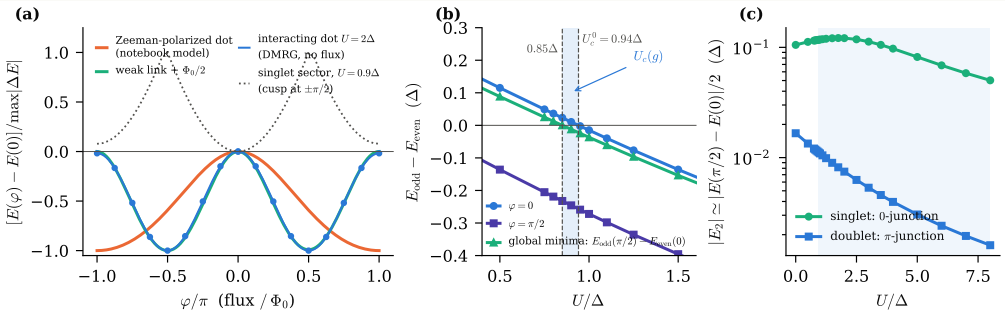
$$- t_J \sum_{\sigma} \left(c^\dagger_{0\sigma} c_{L-1\,\sigma} + \text{h.c.}\right) - t_J \sum_{\sigma} \left(e^{ig(a+a^\dagger)}\, c^\dagger_{1\sigma} c_{0\sigma} + \text{h.c.}\right)$$

- The flux is a Peierls phase on the dot bond; quantized, it becomes the displacement operator $e^{ig(a+a^\dagger)}$ (diamagnetic terms to all orders).
- Parity is conserved: **even** = singlet, a 0 junction; **odd** = Kramers doublet, a π junction (a Cooper pair crosses the singly occupied dot with its spins reversed).

Solved exactly by DMRG on the full electron⊗photon space ($\chi = 300$, Fock cutoff 50–60, sum rule $\langle I \rangle = \langle \Phi \rangle / L$ as acceptance test).

$\Delta = t$, $t_J = 0.5\Delta$, $\epsilon_d = -U/2$, $L = 10$, $\hbar\omega = E_J(2\Delta)/10$.

The junction alone: 0 or π is decided by U

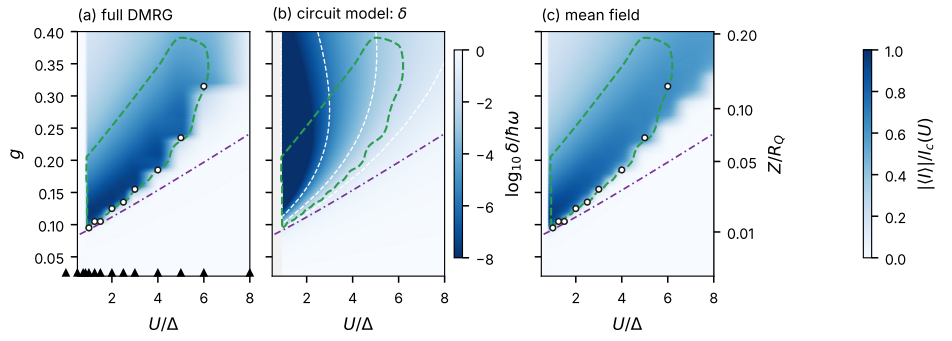


(a) Ring energy vs flux. The doublet at $U = 2\Delta$ is a pure π junction, $E_2 = 7.39 \times 10^{-3}\Delta$; at zero flux it looks exactly like an ordinary weak link biased by $\Phi_0/2$.

(b) Singlet–doublet crossing at $U_c^0 = 0.94\Delta$ (zero flux); the *global* minima cross already at 0.85Δ.
(c) E_J of the π junction falls as $U^{-1.3}$: $E_J/\hbar\omega$ runs from 15 to 2 across the phase diagram.

The phase diagram

Phase diagram in the (U, g) plane – full DMRG



(a) Full DMRG, 427 ground states, seed 10^{-3}: persistent current $|\langle I \rangle|/I_c(U)$. Triangles: computed U; circles: $0.5 I_c$.

(b) Seed-free order parameter: tunnel splitting δ of the circuit model with the DMRG $E(\varphi)$. Grey: singlet 0 loop.

(c) Mean field (product state). Green dashed: $0.5 I_c$ of the circuit model; purple: classical $\beta = 1$. Right axis: Z/R_Q.

How to read it

$U < U_c$: a 0 loop

Singlet ground state, diamagnetic: the mode is *stiffened*, no current. The boundary moves from 0.94Δ to 0.85Δ with g: the zero-point flux itself drives a $0 \to \pi$ switch.

$U > U_c$, $g > g_c(U)$: the condensate

Current $\simeq I_c$, 7–9 photons. g_c follows $\beta = 1$ but lies above it by $\Delta g = 0.02$, 0.04, 0.12 at $E_J/\hbar\omega = 10$, 5, 3.2: zero-point motion opposes the condensation.

$E_J \lesssim 2.5\,\hbar\omega$ ($U \gtrsim 8\Delta$): no order

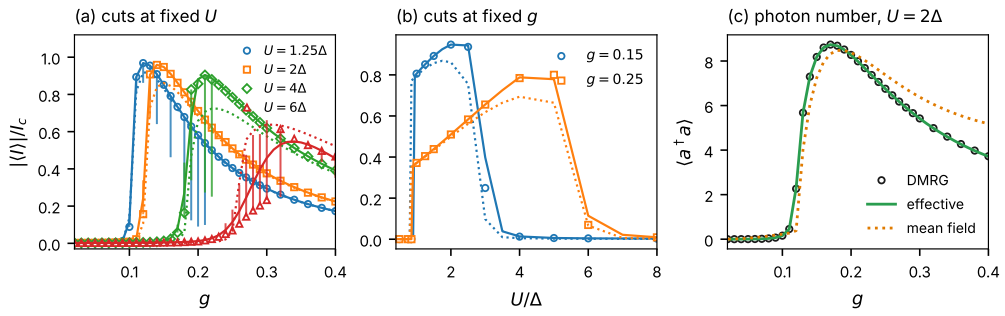
β reaches 2.7, yet the ground state is the cat, $\delta \simeq \hbar\omega$, current $\propto$ seed.

Mean field (c) fails where it matters

Orders at $U = 7$–8Δ with $0.6 I_c$ (absent) and at $U = 0.85\Delta$ with $0.95 I_c$ where the exact state is a 0 loop: a product state cannot be a cat, and replaces the Kramers doublet by a static moment.

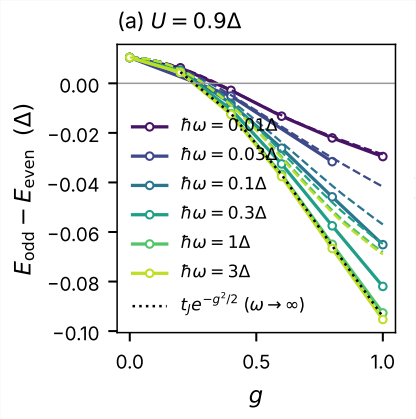
Full DMRG = circuit Hamiltonian to 0.2% of I_c: here the loop *is* a π-fluxonium with the microscopic $E(\varphi)$.

Cuts through the diagram



Symbols: full DMRG; solid: circuit model; dotted: mean field. The sharp step at $U = 1.25,\ 2\Delta$ ($E_J/\hbar\omega \simeq 13,\ 10$) becomes a broad crossover at $U = 6\Delta$ ($E_J/\hbar\omega = 3.2$).

Bonus: the vacuum of the mode switches the junction from 0 to π



$U = 0.9\Delta$: a 0 junction without the mode becomes a π junction at $g \simeq 0.3$ ($Z \simeq 0.6\,\mathrm{k}\Omega$).

- Both parity sectors by full DMRG, no seed, $\hbar\omega = 0.01$–$3\,\Delta$: U_c of the loop moves down by 12% at $g = 0.5$ and 30–40% at $g = 1$.
- Slow mode: zero-point flux costs the singlet energy (its surface curves up) and pays the doublet. Fast mode: the vacuum average $e^{-g^2/2}$ of the Peierls factor reduces the level width (Franck–Condon).
- Observable as the resonator-frequency jump of a gate-tuned dot junction, driven by the impedance alone.

Summary

Summary

1. A π junction sharing its inductance with an LC mode is a π-fluxonium, $H = 4E_C n^2 + \frac{1}{2}E_L\theta^2 + E_J\cos\theta$: for $\beta = 8g^2E_J/\hbar\omega > 1$ a persistent current and a photon condensate appear; the exact ground state is a cat of the two current directions.
2. The full-DMRG phase diagram of the microscopic model (dot + BCS ring + photon) confirms it without approximation: condensate for $U > U_c$, $g > g_c(U)$; threshold pushed above $\beta = 1$ by zero-point motion; no order for $E_J \lesssim 2.5\,\hbar\omega$; mean field fails exactly there and at $U \simeq U_c$.
3. The zero-point flux of the mode is itself a control parameter of the 0–π transition.

Thank you!

Backup: physical units

$$E_J/h \simeq 0.5\,\mathrm{GHz} \times I_c[\mathrm{nA}], \quad \beta = 2\pi(Z/R_Q)(E_J/\hbar\omega)$$

U, Γ, material	E_J/h	I_c	$E_J/10h$	$E_J/10k_B$	$L_{\beta=1}$
2Δ, $\Delta/2$, Al	0.32 GHz	0.65 nA	32 MHz	1.5 mK	510 nH
2Δ, $\Delta/2$, NbTiN	2.1 GHz	4.3 nA	210 MHz	10 mK	76 nH
6Δ, 2Δ, NbTiN	9.9 GHz	20 nA	0.99 GHz	47 mK	17 nH
6Δ, 2Δ, Nb	11.5 GHz	23 nA	1.15 GHz	55 mK	14 nH

- Condensate and cat need $E_J/\hbar\omega \gtrsim 5$ and $\hbar\omega > k_B T$: a strong dot junction in NbTiN/Nb with a $\simeq 1\,\mathrm{GHz}$ mode, $L \simeq 17\,\mathrm{nH}$, $Z \simeq 160\,\Omega$ (full DMRG: 9.6 photons, $0.94 I_c$).
- The flux-driven 0–π switch needs only $\hbar\omega = 0.03$–0.3Δ and $Z = 0.7$–$3\,\mathrm{k}\Omega$: Al, 1–13 GHz, granular-Al superinductors.

Backup: how the current was computed

- Current operator $I = -(e/\hbar)\partial H/\partial\varphi_{\mathrm{cl}}$ through the dot bond, dressed by D; checked against the sum rule $\langle I\rangle = (e/\hbar)(\hbar\omega/2g)\langle a + a^{\dagger}\rangle$ at every point (median violation $10^{-4} I_c$).
- Seed $\varphi_{\mathrm{cl}} = 10^{-3}$ to select a direction; two independent DMRG starts (vacuum side / polarized side) with seed annealing, because the two current states are split only by δ and by $2I^{\star}\varphi_{\mathrm{cl}} \sim 10^{-5}\Delta$.
- Seed 10^{-4}: current unchanged above threshold, reduced $\times 0.09$ below – it is spontaneous.
- $\chi = 400$ changes energies by $< 4 \times 10^{-8}\Delta$ and the current by $< 0.3\%$ away from the crossover.